

Project ID :

25-26J-436

1. Topic (12 words max)

AI Powered Cross Platform Form Generator with Schema Driven Full-Stack Automation

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The persistent challenge of form development in modern web applications arises from redundant implementation of data-entry logic across frontend and backend platforms, leading to validation inconsistencies and increased maintenance effort [1]. Existing solutions such as Formik focus exclusively on React form state management, requiring manual synchronization of validation rules on the server side [2]. JSONForms enables schema-driven UI generation but exhibits significant performance degradation for complex or large forms and lacks server-side code generation capabilities [3] [4]. OpenAPI generators produce API stubs independently for client or server, yet they fail to address form-specific validation rules, UI layout needs, and full-stack consistency [5].

This duplication of effort not only increases the risk of mismatches in validation logic but also prolongs development cycles and hinders productivity [6]. The absence of a unified schema driven system that can simultaneously generate both frontend form components and backend data models with coherent validation has been widely noted in literature on schema-driven development and model-driven architecture [7] [8]. Security and compliance requirements, such as GDPR and HIPAA, further complicate form implementation by demanding consistent enforcement of data privacy rules across all layers of the application stack [9].

Another important research problem is that even when model-driven or schema-driven approaches are used to generate forms, many existing tools require specialized technical skills and are not accessible to all developers or end-users. Studies highlight that traditional model-based UI generation often involves complex transformation processes, multiple meta-models, and specialized languages, which can be challenging for junior developers or teams without deep expertise. As a result, productivity gains from automation are limited, and small companies or new teams may struggle to adopt these technologies effectively. Furthermore, the lack of user-friendly, intelligent interfaces for schema or model creation restricts the broader adoption of automated form generation. This gap shows the need for simplified, AI-assisted tools that allow both technical and non-technical users to easily define, validate, and deploy forms while ensuring consistency and compliance across platforms.

References

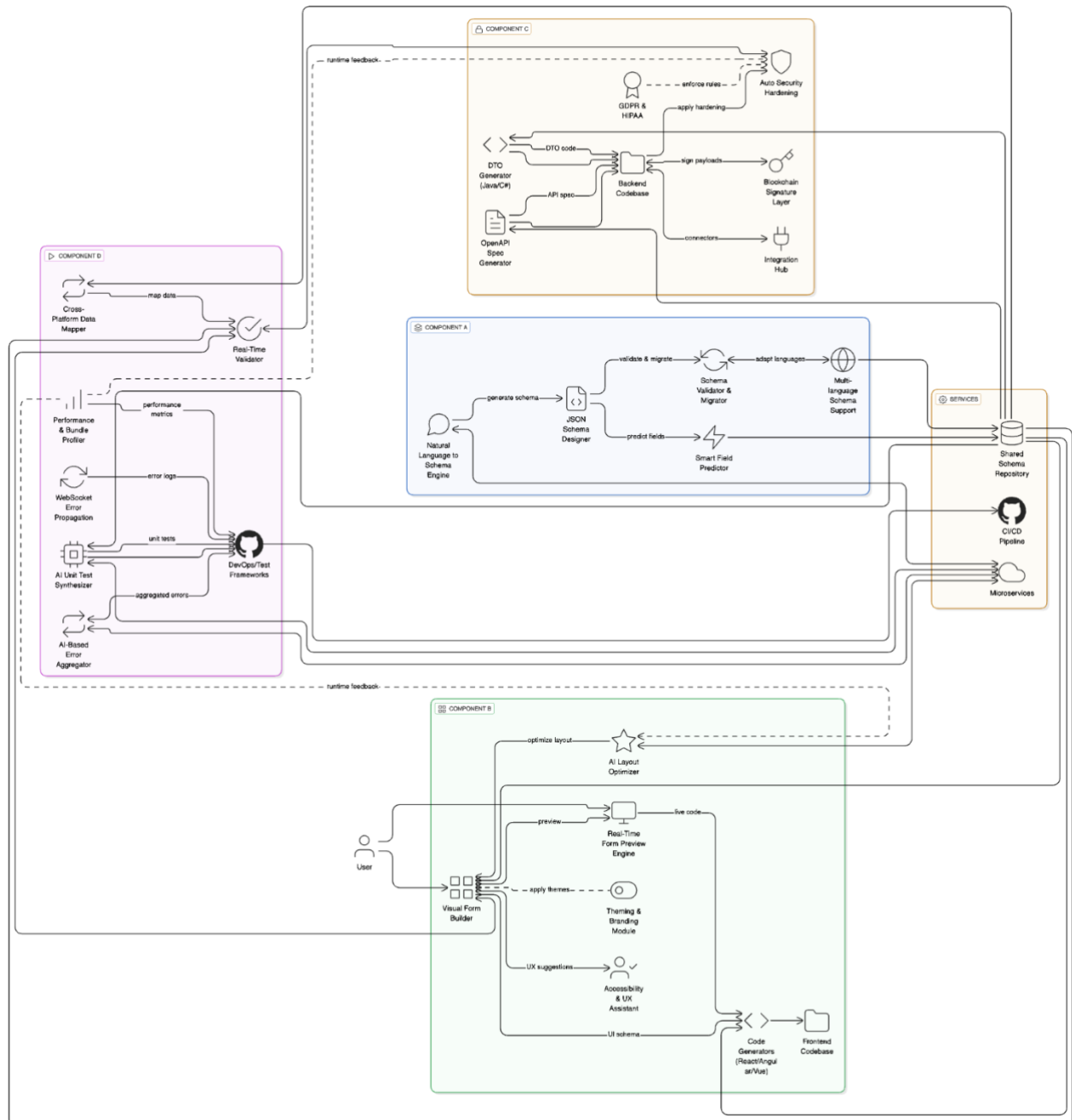
- [1] M. Fowler, *Patterns of Enterprise Application Architecture*, Boston, MA, USA: Addison-Wesley Professional, 2013.
- [2] F. D. Team, "Formik: Build forms in React, without tears,," 2023. [Online]. Available: <https://formik.org/docs/overview>.
- [3] J. Contributors, "JSONForms: Performance and Limitations," 2023. [Online]. Available: <https://jsonforms.io/>.
- [4] A. Osmani, *Bundle Size Optimization in Modern Web Applications*, 2017.
- [5] O. Initiative, "OpenAPI Specification v3.1.0," 2023. [Online]. Available: <https://spec.openapis.org/>.
- [6] J. Nielsen and H. Loranger, *Prioritizing Web Usability*, Berkeley, CA, USA: New Riders Press, 2006.
- [7] M.-J. P. S. a. J. H. H. D. C. Schmidt, "Model-Driven Software Development," *IEEE Comput*, vol. 39, no. 10, p. 28–34, Oct. 2006.
- [8] J. B. a. W. L. S. A. Kleppe, *MDA Explained: The Model Driven Architecture*, Boston, MA, USA: Addison-Wesley, 2003.
- [9] GDPR.eu, "GDPR Compliance Guidelines for Software Development," 2023. [Online]. Available: <https://gdpr.eu/>.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

A unified schema-driven platform is proposed that enables developers to define form structures, validation rules, and UI metadata once and generate consistent code for multiple frontend and backend frameworks without redundancy.

The solution architecture comprises four modules

1. Intelligent Schema Definition & AI Integration, which leverages JSON/YAML, drag-and-drop, and natural language inputs with AI-assisted schema generation and validation
2. Multi-Platform Frontend Generator & Visual Builder, which produces React, Angular, and Vue components with built-in validation and a visual design interface
3. Backend Generator & Security Integration, which outputs secure Spring Boot and ASP.NET Core code, complete with validation annotations, OpenAPI specifications, and compliance modules
4. Runtime Engine & Advanced Testing, which implements a shared validation engine, real-time error reporting, type-safe data binding, and AI-powered automated test case generation integrated into CI/CD pipelines.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise

- Advanced knowledge in software engineering, particularly in cross-platform development (React, Angular, Spring Boot, .NET).
- Expertise in artificial intelligence, including natural language processing for schema generation and machine learning for predictive analytics.
- Security engineering proficiency (OWASP standards, GDPR/HIPAA compliance) and accessibility standards (WCAG 2.1).

Knowledge

- Understanding of template-based code generation for multi-framework support.
- Knowledge of real-time communication protocols (WebSocket).
- Validation synchronization across JavaScript, Java, and C# is necessary.
- Familiarity with test automation frameworks (Jest, JUnit, xUnit).

Data Requirements

- Form Schemas: Existing JSON/YAML form definitions for training AI models.
- Validation Rules: Datasets of frontend/backend validation logic for consistency testing.
- User Interaction Logs: Anonymized form-completion data to train UX optimization models.

8. Objectives and Novelty

Main Objective Develop an AI-powered cross-platform form generation system that eliminates validation inconsistencies between frontend and backend while supporting multiple programming languages through intelligent schema-driven code generation.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Rajapakshe P H Y L IT22305350	Intelligent Schema Definition & AI Integration	• Design unified JSON/YAML schema format with validation rules • Implement natural language to schema conversion using LLMs • Develop smart field prediction algorithms • Create schema versioning and migration system	• First conversational schema generation interface • Context-aware field suggestion engine • Multi-language schema translation capabilities • AI-powered schema pattern recognition
Gunawardena T S IT22350114	Multi-Platform Frontend Generator & Visual Builder	• Build React/Angular/Vue code generators with validation • Create drag-and-drop visual form builder • Implement AI layout optimization system • Generate TypeScript interfaces automatically	• Multi-modal form interfaces (voice/gesture) • AI-driven responsive design suggestions • WCAG 2.1 compliance automation • Real-time UX completion analytics

Dissanayake D M T V IT22003546	Backend Generator & Security Integration	• Generate Spring Boot/.NET DTOs with validation annotations • Implement OWASP security vulnerability scanning • Create GDPR/HIPAA compliance automation • Build OpenAPI specification generator	• Quantum-safe encryption integration • Blockchain-based form authenticity verification • Automated penetration testing for generated code • Enterprise system integration hub
Liyanage M I H IT22332608	Runtime Engine & Advanced Testing	• Develop AI-powered unit test synthesis for all platforms • Create cross-platform data binding engine • Implement real-time validation synchronization via WebSockets • Build performance benchmarking system	• LLM-generated test cases with edge case detection • Intelligent error orchestration with plain English translation • Coverage guided test optimization (>80%) • Real-time validation consistency across JS/Java/C#

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Rajapakshe P H Y L IT22305350	Specialized expertise in artificial intelligence and natural language processing is applied to design an advanced schema definition engine. AI models are integrated for natural language to schema conversion, smart field prediction, and schema optimization. Compliance with software engineering specialization is ensured by implementing multi-language schema support, schema validation, and versioning, aligning with best practices in intelligent software systems.
Gunawardena T S IT22350114	Advanced knowledge in cross-platform frontend engineering is demonstrated by developing code generators for React, Angular, and Vue, and building a visual drag-and-drop form builder. Machine learning is used for AI-powered layout optimization and accessibility compliance. The component complies with the specialization by enabling multi-framework UI generation, adaptive interfaces, and automated accessibility checks, reflecting modern frontend and UX engineering standards.
Dissanayake D M T V IT22003546	Security engineering proficiency and backend development skills are used to generate secure, validated DTOs for Spring Boot and .NET, and to automate OpenAPI documentation. Security and compliance frameworks (OWASP, GDPR, HIPAA) are integrated into code generation. The component aligns with the specialization by incorporating advanced security controls, automated compliance checks, and enterprise backend integration, ensuring robust and compliant backend systems.
Liyanage M I H IT22332608	Expertise in automated testing, data binding, and real-time systems is applied to build an AI-powered test generation engine, cross-platform data binding, and unified validation logic. Real-time error propagation and performance benchmarking are implemented using WebSockets and profiling tools. The component complies with the specialization by ensuring validation consistency, high test coverage, and robust runtime analytics across platforms, reflecting advanced software quality and runtime engineering practices.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Jeewaka	Perera	
Co-Supervisor				
External Supervisor	N/A	N/A	N/A	N/A
Summary of external supervisor's (if any) experience and expertise				

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.